

Kanha Batra

Email: kanhabatra2008@gmail.com | | LinkedIn: kanhabatra | Scholar: Google Scholar

SUMMARY

Stanford Postdoctoral Scholar (Giocomo Lab) with a Ph.D. in Machine Learning from UC San Diego (advisors: Kay Tye, Terrence Sejnowski). Co-first author on two **Nature** papers; 7+ peer-reviewed publications spanning neural network architectures, probabilistic models, and medical imaging ML. Core expertise in continual and meta-learning, deep reinforcement learning, Bayesian inference, and large-scale neural data modeling. Industry experience at Amazon Science.

EDUCATION

University of California, San Diego

Ph.D. in Machine Learning & Data Science

Electrical & Computer Engineering

June 2020 – June 2024

University of California, San Diego

M.S. in Machine Learning & Data Science

Electrical & Computer Engineering

September 2018 – June 2020

Indian Institute of Technology, Delhi

B.Tech. in Chemical Engineering

Chemical Engineering

July 2014 – May 2018

SKILLS

- **Frameworks & Libraries:** PyTorch, TensorFlow, JAX, Hugging Face, scikit-learn, NumPy, SciPy, pandas, wandb
- **Methods:** Continual/lifelong learning, meta-learning, deep reinforcement learning, transformer architectures, probabilistic graphical models (HMM, GLM), variational inference, Bayesian optimization, SHAP/XAI, multi-task learning
- **Infrastructure:** Python, AWS, Spark, Git, Docker, SLURM/HPC clusters, MATLAB, Linux

EXPERIENCE

Giocomo Lab — Advisors: Lisa Giocomo & Scott Linderman

Postdoctoral Fellow

Stanford University

September 2024 – Present

- Developing biologically-grounded neural network architectures for continual learning, using hippocampal remapping as a computational framework for non-stationary representation learning in AI systems.
- Applying state-space models to large-scale neural population recordings in collaboration with the Linderman lab, targeting scalable inference for latent dynamics in high-dimensional datasets.

Systems Neuroscience Lab — Advisors: Kay Tye & Terrence Sejnowski

Graduate Student Researcher

Salk Institute, UC San Diego

July 2019 – August 2024

- Built an unsupervised probabilistic state-space model (Hidden Markov Model coupled with Generalized Linear Models) to decode latent behavioral states from high-dimensional prefrontal cortex population recordings during social competition; results constituted a key computational contribution to a **co-first-author Nature (2022)** paper.
- Designed a neuromodulation-inspired **mixture-of-experts architecture** as a meta-learning framework for continual multi-task learning across non-stationary environments; validated via deep reinforcement learning benchmarks and in-vivo neuroscience experiments; presented at Cosyne 2024 and SfN 2023.
- Contributed probabilistic decoding analysis to the **Nature (2022)** study on neurotensin and valence assignment in the amygdala (Li et al.).

Seller Partner Recruitment & Success Team — Amazon Science

Applied Science Intern

Amazon HQ, Seattle

June 2022 – September 2022

- Implemented Shapley Additive Explanations (SHAP) to quantify feature importance and identify high-impact events in temporal classification tasks on recurrent neural networks, improving model interpretability for seller recruitment decisions.
- Pipeline operated at production scale; findings directly informed feature engineering strategy for the downstream seller scoring model.

Multimodal Imaging Lab — Advisors: Tyler Seibert & Anders Dale

Graduate Student Researcher

ACTRI, UC San Diego

December 2018 – June 2020

- Developed an ML classifier (restriction spectrum imaging + supervised learning) to characterize and segment bone metastases in prostate cancer patients; work contributed to two peer-reviewed publications in *Scientific Reports* (2022) and *Journal of Magnetic Resonance Imaging* (2021).

SELECTED PUBLICATIONS

- Nancy Padilla-Coreano*, **Kanha Batra***, Makenzie Patarino, Zexin Chen, et al. & Kay M. Tye. Cortical ensembles orchestrate social competition through hypothalamic outputs. **Nature, 2022.** (*co-first authors)
- Hao Li*, Praneeth Namburi*, Jacob M. Olson*, ..., **Kanha Batra**, ..., Kay M. Tye. Neurotensin orchestrates valence assignment in the amygdala. **Nature, 2022.**
- Austin A. Coley, **Kanha Batra**, Jeremy M. Delahanty, et al. & Kay M. Tye. Predicting Future Development of Stress-Induced Anhedonia From Cortical Dynamics and Facial Expression. **Preprint, 2024.**
- Chris A. Leppla, Laurel R. Keyes, Gordon Glover, ..., **Kanha Batra**, ..., Kay M. Tye. Thalamus sends information about arousal but not valence to the amygdala. **Psychopharmacology, 2023.**
- Christine H. Feng, Christopher C. Conlin, **Kanha Batra**, Ana Rodriguez-Soto, et al. & Tyler M. Seibert. Voxel-level Classification of Prostate Cancer on MRI: Improving Accuracy Using Four-Compartment Restriction Spectrum Imaging. **Journal of Magnetic Resonance Imaging, 2021.**